

Na'muna uchun misollar yechimi

1-misol. R^m fazodagi ushbu

$$B_r(a) = \{x \in R^m : \rho(x, a) < r\}$$

sharning ochiq to'plam ekanligi ko'rsatilsin.

◀ $\forall x^0 \in B_r(a)$ nuqtani olamiz. Unda

$$r - \rho(x^0, a)$$

miqdor musbat bo'ladi. Uni δ deylik: $\delta = r - \rho(x^0, a)$. (22-chizma)

Endi x^0 nuqtaning ushbu

$$U_\delta(x^0) = \{x \in R^m : \rho(x, x^0) < \delta\}$$

atrofini qaraymiz.

Bunda $U_\delta(x^0) \subset B_r(a)$ bo'ladi. Haqiqatdan ham,

$$\forall x \in U_\delta(x^0) \Rightarrow \rho(x, x^0) < \delta$$

bo'lib, masofaning 3)-xossasiga ko'ra

$$\rho(x, a) \leq \rho(x, x^0) + \rho(x^0, a) < \delta + \rho(x^0, a) = r$$

bo'ladi. Demak,

$$\forall x \in U_\delta(x^0) \Rightarrow x \in B_r(a)$$

Bundan $U_\delta(x^0) \subset B_r(a)$ bo'lishi kelib chiqadi.

Demak, $B_r(a)$ to'plamning har bir nuqtasi uning ichki nuqtasi bo'ladi. Binobarin, $B_r(a)$ ochiq to'plam. ▶

Misollar

1. Agar $G_1 \subset R^m$, $G_2 \subset R^m$ ochiq to'plamlar bo'lsa,

$$G_1 \cup R^m, G_2 \cap R^m$$

to'plamlarning ochiq to'plam bo'lishi ko'rsatilsin.

2. Agar $F_1 \subset R^m$, $F_2 \subset R^m$ yopiq to'plamlar bo'lsa,

$$F_1 \cup R^m, F_2 \cap R^m$$

to'plamlarning yopiq to'plam bo'lishi ko'rsatilsin.

3. Aytaylik (X, ρ) metrik fazo berilgan bo'lsin, $x_0 \in X$ va $r > 0$.
 $B_r(a) = \{x \in R^m : \rho(x, a) < r\}$ ochiq shar va $C := \{x \in X : \rho(x, x_0) \leq r\}$ yopiq shar
berilgan.

a) $\bar{B} \subseteq C$ isbotlang.

b) $\bar{B} \subseteq C$ ga teng bo'lmagan holat uchun $r > 0$ radiusli (X, ρ) metrik fazoga misollar
keltiring.

Test

1. Agar x^0 nuqta G to'plamga tegishli bo'lgan $U_\varepsilon(x^0)$ atrofga ega bo'lsa, $(U_\varepsilon(x^0) \subset G)$ x^0 nuqta G to'plamning nuqtasi deyiladi.
 A) ichki B) chegaraviy C) limit D) urinish
2. Agar x^0 nuqtaning ixtiyoriy $U_\varepsilon(x^0)$ atrofida $(\forall \varepsilon > 0)$ F to'plamning x^0 dan farqli kamida bitta nuqtasi bo'lsa, x^0 nuqta F to'plamning nuqtasi deyiladi.
 A) limit B) chegaraviy C) ichki D) urinish
3. Agar x^0 nuqtaning ixtiyoriy $U_\varepsilon(x^0)$ atrofida ham M to'plamning, ham $R^m \setminus M$ to'plamning nuqtalari bo'lsa, x^0 nuqta M to'plamning nuqtasi deyiladi.
 A) chegaraviy B) limit C) ichki D) urinish
4. G to'plamning har bir nuqtasi uning ichki nuqtasi bo'lsa, u deyiladi.
 A) ochiq to'plam B) yopiq to'plam C) chegara D) ochiq shar
5. Agar $F \subset R^m$ to'plamning barcha limit nuqtalari shu to'plamga tegishli bo'lsa, F deyiladi.
 A) yopiq to'plam B) ochiq to'plam C) chegara D) ochiq shar
6. x va y nuqtalar orasidagi masofa to'g'ri keltirilgan qatorni ko'rsating.
 A) $\sqrt{\sum_{k=1}^m (y_k - x_k)^2}$ B) $\sqrt{\sum_{k=1}^m (y_k + x_k)^2}$ C) $\sum_{k=1}^m (y_k - x_k)^3$ D) $\sum_{k=1}^m (y_k - x_k)^2$
7. $x^0 \in R^m$ nuqtaning sferik atrofni toping.
 A) $U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) < \varepsilon\}$ B) $U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) \leq \varepsilon\}$
 C) $U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) > \varepsilon\}$ D) $U_\varepsilon(x^0) = \{x \in R^m : \rho(x, x^0) \geq \varepsilon\}$
8. Agar $G_1 \subset R^m$, $G_2 \subset R^m$ ochiq to'plamlar bo'lsa, u holda quyidagi to'plamlar ichida ochiq to'plamlarni ko'rsating.
 A) $G_1 \cup R^m$, $G_2 \cap R^m$ B) $G_1 \setminus R^m$, $G_2 \setminus R^m$ C) $G_1 \Delta R^m$, $G_2 \Delta R^m$
 D) $G_1 \Delta R^m$, $G_2 \setminus R^m$

9. Agar $F_1 \subset R^m, F_2 \subset R^m$ yopiq to'plamlar bo'lsa, u holda quyidagi to'plamlar ichida yopiq to'plamlarni ko'rsating.

A) $F_1 \cup R^m, F_2 \cap R^m$

B) $F_1 \setminus R^m, F_2 \setminus R^m$

C) $F_1 \Delta R^m, F_2 \Delta R^m$

D) $F_1 \Delta R^m, F_2 \setminus R^m$

10. M to'plamning chegarasi to'g'ri belgilangan qatorni qatorni ko'rsating.

A) $\partial(M)$

B) $\overline{M}$

C) M'

D) $\text{int}(M)$